

MEDHA Behaviour Engine

Supporting Research Literature

Architecture & methodology references, and the clinical evidence base behind each design decision

Compiled to accompany the MEDHA Behaviour Engine prototype for Smart India Hackathon review. Every entry below was verified via live web search and links to a real, accessible source (arXiv, PubMed/PMC, an open-access journal, or the original publisher). Where full-text access may require an institutional login rather than being freely downloadable, that is noted explicitly.

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Part A — Architecture & Methodology References

These establish the technical foundations the Behaviour Engine is built on: the reference platform its data structure was modeled after, the algorithms actually used or planned, and the methodology for synthetic-data-first development.

A1. Digital phenotyping foundation

A1.1. Beiwe: A data collection platform for high-throughput digital phenotyping

Onnela JP, Dixon C, Griffin K, et al. (2021). *Journal of Open Source Software*, 6(68), 3417.

<https://doi.org/10.21105/joss.03417>

Open access. The platform paper for Beiwe — the reference system the Behaviour Engine's raw-event structure (survey timing, answer interaction fields) was mapped from before being adapted to MEDHA's own schema.

A1.2. New Tools for New Research in Psychiatry: A Scalable and Customizable Platform to Empower Data Driven Smartphone Research

Torous J, Kiang MV, Lorme J, Onnela JP (2016). *JMIR Mental Health*, 3(2), e16.

<https://doi.org/10.2196/mental.5165>

Open access. The original Beiwe introduction paper.

A1.3. Digital Phenotyping: Data-Driven Psychiatry to Redefine Mental Health

Journal of Medical Internet Research (2023), 25, e44502.

<https://www.jmir.org/2023/1/e44502>

Open access. Conceptual framing for using continuous behavioural data as a psychiatric signal.

A1.4. Digital Phenotyping for Monitoring Mental Disorders: Systematic Review

PMC (2023).

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10753422/>

Open access. Systematic review specifically on relapse/exacerbation prediction — the closest published framing to the engine's `future_deterioration_7d` target.

A2. Personal-baseline / idiographic modeling

A2.1. Digital phenotyping of affect and stress in emerging adults

Frontiers in Digital Health (2026).

<https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2026.1799541/full>

Open access. Directly compares idiographic (within-person) vs. nomothetic (population) models for affective outcomes — the same design choice the Behaviour Engine makes in favour of personal baselines.

A3. Anomaly detection (Isolation Forest)

A3.1. Isolation Forest

Liu FT, Ting KM, Zhou ZH (2008). *Proc. 8th IEEE International Conference on Data Mining (ICDM)*, 413-422.

https://www.researchgate.net/publication/224384174_Isolation_Forest

Freely downloadable PDF. The original algorithm underlying the currently deployed anomaly-scoring model.

A3.2. Isolation-Based Anomaly Detection

Liu FT, Ting KM, Zhou ZH (2012). *ACM Transactions on Knowledge Discovery from Data*, 6(1), 1-39.

https://www.researchgate.net/publication/239761771_Isolation-Based_Anomaly_Detection

Freely downloadable PDF (ACM version is subscription-gated). Extended journal treatment with full theoretical analysis.

A4. Sequence modeling & gated multimodal fusion

A4.1. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation

Cho K, van Merriënboer B, Gulcehre C, et al. (2014). arXiv:1406.1078.

<https://arxiv.org/abs/1406.1078>

Open access. The paper introducing the GRU cell planned for the fusion sequence model.

A4.2. Gated Multimodal Units for Information Fusion

Arevalo J, Solorio T, Montes-y-Gomez M, Gonzalez FA (2017). arXiv:1702.01992.

<https://arxiv.org/abs/1702.01992>

Open access. The canonical reference for the gating mechanism planned for late fusion of behaviour, text, and voice modalities.

A4.3. Multimodal Machine Learning: A Survey and Taxonomy

Baltrusaitis T, Ahuja C, Morency LP (2017/2019). *IEEE TPAMI*, 41(2), 423-443. arXiv:1705.09406.

<https://arxiv.org/abs/1705.09406>

Open access. Standard taxonomy of early/late/hybrid fusion strategies, useful for justifying gated late fusion as a deliberate choice.

A4.4. Multimodal Machine Learning in Mental Health (survey)

arXiv:2407.16804 (2024).

<https://arxiv.org/pdf/2407.16804>

Open access. Domain-specific survey with gated-fusion depression-detection examples and DAIC-WOZ benchmark F1 scores to compare against.

A4.5. Multimodal Sensing for Depression Risk Detection: Integrating Audio, Video, and Text Data

PMC (2024).

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11207438/>

Open access. Closest existing published architecture to the planned behaviour + text + voice fusion.

A5. Real-time crisis / suicide-risk monitoring

A5.1. Digital phenotyping of suicidal thoughts

Kleiman EM, Turner BJ, Fedor S, Beale EE, Picard RW, Huffman JC, Nock MK (2018). *Depression and Anxiety*, 35(7), 601-608.

<https://kleimanlab.org/wp-content/uploads/2018/11/Kleiman-et-al.-2018-Digital-Phenotyping-of-Suicidal-Thoughts.pdf>

Freely downloadable PDF via the author's lab site. Closest published analogue to the engine's real-time, app-based risk-monitoring target.

A5.2. Don't Miss the Moment: A Systematic Review of Ecological Momentary Assessment in Suicide Research

PMC (2022).

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9120419/>

Open access. EMA methodology review directly relevant to the check-in completion/timing feature design.

A6. Synthetic data generation methodology

A6.1. A Multifaceted Benchmarking of Synthetic Electronic Health Record Generation Models

Nature Communications (2022), 13, 7609.

<https://www.nature.com/articles/s41467-022-35295-1>

Open access. Real utility/privacy benchmarking framework adaptable for the project's own data-card validation checklist.

A6.2. Reliable generation of privacy-preserving synthetic electronic health record time series via diffusion models

Journal of the American Medical Informatics Association / *PMC* (2024).

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11491591/>

Open access. Specifically addresses longitudinal time-series synthetic generation, matching the 30-day candidate trajectory design.

A7. Alert design & explainability

A7.1. Effects of workload, work complexity, and repeated alerts on alert fatigue in a clinical decision support system

Ancker JS, et al. (2017). *BMC Medical Informatics and Decision Making*, 17, 36.

<https://bmcmmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-017-0430-8>

Open access. Real-world evidence on false-positive alert rates in deployed clinical systems — useful context for the engine's own precision/recall trade-off at the triage threshold.

A7.2. Explainable artificial intelligence for mental health through transparency and interpretability for understandability

npj Digital Medicine (2023), 6, 6.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9849399/>

Open access. Argues for a mental-health-specific definition of explainability — matches the gate-weights-as-explanation design planned for the fusion layer.

A7.3. Explainable AI: A Review of Machine Learning Interpretability Methods

Linardatos P, Papastefanopoulos V, Kotsiantis S (2020). *Entropy*, 23(1), 18.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7824368/>

Open access. General reference for SHAP/LIME-style methods for building explanation surfaces on top of the trained models.

Part B — Clinical Evidence Base

Organized by which specific design decision each entry supports — the framing to use when a judge asks “why did you build it this way,” not a generic reading list.

B1. Why the engine is case-aware

B1.1. The mental health of crime victims: Impact of legal intervention

Herman JL (2003). *Journal of Traumatic Stress*, 16(2), 159-166.

<https://doi.org/10.1023/A:1022847223135>

Foundational review establishing that legal-system involvement carries real, documented mental health risk for victims — direct clinical justification for treating case_stage and case_event_today as first-class features.

B1.2. Secondary victimization of rape victims: insights from mental health professionals who treat survivors of violence

Campbell R, Raja S (1999). *Violence and Victims*, 14(3), 261-275.

<https://pubmed.ncbi.nlm.nih.gov/10606433/>

The “second rape” / secondary victimization literature specifically for sexual violence survivors navigating investigation and prosecution.

B1.3. Do Trials of Perpetrators Retraumatize Crime Victims?

Orth U, Maercker A (2004). *Journal of Interpersonal Violence*, 19(2), 212-227.

<https://pubmed.ncbi.nlm.nih.gov/15006002/>

A longitudinal study of 31 crime victims tracking PTSD reactions across a trial.

Honest caveat, worth citing deliberately: this specific study did NOT find support for the retraumatization-by-trial hypothesis in its sample. Presenting this alongside B1.1-B1.2 is stronger than only citing confirming evidence — it shows the case-aware design accounts for genuine complexity in the evidence (which is exactly why case_change_score is built as a noisy, bidirectional signal rather than a fixed “trial = harm” rule).

B2. Clinical validation of the idiographic-baseline principle

B2.1. Differential temporal utility of passively sensed smartphone features for depression and anxiety symptom prediction: a longitudinal cohort study

npj Mental Health Research (2024), 3, 1.

<https://www.nature.com/articles/s44184-023-00041-y>

Open access. **Strongest citation in this document.** Real clinical cohort (n=1,013); XGBoost achieved AUC 0.906 predicting symptom-severity-band crossings, and each person's own typical symptom level was the only statistically significant predictor beyond the most recent score — without it, the most consequential worsening transitions went undetected. Independent clinical confirmation of the exact personal-baseline premise the Behaviour Engine is built on, with an AUC essentially matching this project's own 0.903.

B3. Passive engagement as a validated relapse marker

B3.1. Digital phenotyping for predicting relapse in psychiatric disorders: a systematic review of passive sensing approaches

BMC Psychiatry (2026).

<https://link.springer.com/article/10.1186/s12888-026-08157-z>

Open access. Reviews the relapse-prediction literature that `future_deterioration_7d` is modeled after.

B3.2. From smartphone data to clinically relevant predictions: A systematic review of digital phenotyping methods in depression

Comprehensive Psychiatry / ScienceDirect (2024).

<https://www.sciencedirect.com/science/article/pii/S0149763424000095>

May require institutional access. Reports real sensitivity/specificity/AUC ranges from multiple studies (e.g. 0.70/0.65/0.75 and 0.70/0.77/0.79) — useful benchmark figures to place next to this project's own numbers.

B3.3. Exploring the Effects of In-App Components on Engagement With a Symptom-Tracking Platform Among Participants With Major Depressive Disorder (RADAR-Engage)

PMC, protocol for the RADAR-MDD cohort study (n=623 across 3 sites).

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8734922/>

Open access. A funded, multi-site clinical cohort built specifically around engagement with a symptom-tracking app as a clinical variable — direct precedent for treating `engagement_deviation` as meaningful in its own right.

B4. Missed check-ins as clinical signal, not noise

B4.1. Not missing at random: Missing data are associated with clinical status and trajectories in an electronic monitoring longitudinal study of bipolar disorder

Journal of Psychiatric Research / ScienceDirect (2024).

<https://www.sciencedirect.com/science/article/pii/S0022395624002425>

May require institutional access. Real clinical evidence that WHEN someone stops responding is itself associated with their clinical trajectory — the direct clinical justification for treating `inactivity_score` as a feature rather than something to impute away.

B4.2. Readdressing the Ongoing Challenge of Missing Data in Youth Ecological Momentary Assessment Studies: Meta-Analysis Update

Journal of Medical Internet Research (2025), 27, e65710.

<https://www.jmir.org/2025/1/e65710>

Open access. Establishes compliance/dropout as a studied clinical variable with real benchmark rates (e.g. 48% average dropout in smartphone MDD interventions) — context for why `timing_observed` / the GRAY insufficient-history state matters operationally, not just as an edge case.

Part C — Component-to-Literature Map

Quick reference for defending a specific design decision to a judge.

Engine component / decision	Backed by
Case-aware features (case_stage, case_event_today, case_charge_score)	B1.1, B1.2, B1.3
Personal-baseline z-scores instead of population norms	A2.1, B2.1
Isolation Forest for anomaly scoring	A3.1, A3.2
engagement_deviation as a standalone score	B3.3, B2.1
inactivity_score / missed-checkin treatment	B4.1, B4.2
future_deterioration_7d target design	A1.4, B3.1
Real-time / EMA-style check-in monitoring	A5.1, A5.2, B4.2
Planned GRU + gated late fusion architecture	A4.1, A4.2, A4.3, A4.4, A4.5
Synthetic-data-first development approach	A6.1, A6.2
0.50 triage threshold / precision-recall trade-off	A7.1
Gate weights as the explanation surface for alerts	A7.2, A7.3

Note on evidentiary status: the papers above establish that the underlying design principles (idiographic baselines, engagement as a marker, missingness as signal, legal-process trauma risk) are clinically recognized and, in several cases, independently validated with comparable predictive performance to this project's own results. They do not constitute clinical validation of the MEDHA Behaviour Engine itself — this remains a synthetic-data development benchmark, not a validated diagnostic or predictive instrument, until tested against real MEDHA data.